

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Software engineer with Java 21, TypeScript, and Python in production -- delivering 394 commits and 7 production deployments at Isomer AI across backend features, REST APIs, and full SDLC ownership. Built a transactional Java 21 + Spring Boot inventory service with concurrency-safe PostgreSQL locking, Testcontainers integration tests, and GitHub Actions CI/CD. NC State MS CS (May 2026, Raleigh) with Operating Systems coursework; coding-interview ready across all three Veeva EDP languages.

CORE COMPETENCIES

Java 21 + Spring Boot

TypeScript + Node.js

Python + REST APIs

PostgreSQL + SQL

Spring Security + RBAC

Docker + CI/CD

Testcontainers + Testing

Agile SDLC

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Owned end-to-end SDLC for 10+ backend features on a production AI platform -- **394 commits, 7 production deployments**, and PR reviews across the engineering team, tracking monthly release cadence
- Built TypeScript + Node.js + Express REST APIs with role-based access control (Auth0), Temporal.io workflow orchestration, real-time event pipelines (Pusher), and PostgreSQL + Redis + Prisma data layer
- Implemented document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage) and integrated multi-model AI tooling across Claude, GPT-4o, and Gemini with MCP, Docker, and Sentry

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Designed a cloud-based genomic data platform with JBrowse 2 enabling interactive exploration of 500GB datasets for 100+ researchers worldwide, deployed on Google Cloud Run
- Applied hyperparameter tuning and time-series analysis to ML models, improving accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated monitoring dashboards with Grafana and supported model validation

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a data-driven support ticket analysis system, reducing processing time by 15%
- Implemented distributed data pipelines (Hadoop + Spark + Kafka), reducing ticket-processing latency by 20%

PROJECTS

Inventory Reservation Service

Java 21 + Spring Boot

github.com/Aditya-k24/InventoryReservationService

Built a transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent POST endpoints with RFC 9457 error responses and automated expiry via scheduled cleanup job. Verified concurrency correctness with Testcontainers integration tests -- two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, Spring Security (role-based), multi-stage Docker build, GitHub Actions CI/CD.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Data JPA, Spring Security, Testcontainers, Docker, GitHub Actions

AlgoPulse -- Agentic AI Platform TypeScript + Temporal github.com/Aditya-k24/AlgoPulse

Architected an event-driven AI orchestration platform; identified API-polling bottleneck under load mid-build and pivoted to Supabase Edge Functions streaming AI reasoning loops directly to the client. Scaled with Temporal.io and Kafka for state synchronization and fault tolerance, handling 2,000 async inference requests/second without state loss -- demonstrating full Agile iteration cycle from design through production rethink.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank -- Real-Time Leaderboard Platform TypeScript + NestJS github.com/Aditya-k24/PulseRank

Built an event-driven leaderboard and activity feed: MySQL for transactional event writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads, live updates over Server-Sent Events. Validated multi-store consistency with Testcontainers integration tests; CI/CD with GitHub Actions.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, GitHub Actions, OpenTelemetry, Prometheus

VectorLift -- Semantic Search Engine Python + Kafka github.com/Aditya-k24/VectorLift

Built a production-grade semantic search system: Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI search server with per-stage latency tracking; evaluated 6 pipeline configurations (BM25, dense retrieval, cross-encoder re-ranking) with NDCG@10, MRR@10, MAP, and bootstrap significance testing.

Python, FastAPI, Kafka, Elasticsearch, Qdrant, PostgreSQL, Redis, Prometheus, Grafana, Docker Compose

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Raleigh, NC | Courses: Operating Systems, Artificial Intelligence, Software Engineering, Algorithms, Data Science, Object Oriented Development

Mumbai University -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, OS, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

Languages: Java, TypeScript, Python, JavaScript, SQL **Backend:** Spring Boot, Node.js, Express, NestJS, FastAPI, REST APIs
Data: PostgreSQL, MySQL, MongoDB, Redis, Flyway, Prisma, Spring Data JPA **DevOps:** Docker, GitHub Actions, CI/CD, GCP Cloud Run, AWS
Testing: Testcontainers, integration testing, k6 benchmarking **Other:** Temporal.io, Kafka, Auth0, Prometheus, Grafana, Git